

Historical (1850–1995) nitrogen changes in UK catchments recorded by lake sediment $\delta^{15}\text{N}$

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Environmental context. Unprecedented modification of the global nitrogen cycle by human activity is strongly imprinted in UK lakes by the changing abundance of nitrogen isotopes found in layers of lake sediment. UK lakes in the last century have responded to this change depending on their location, particularly if they are remote and occur above ~300-m altitude. Historical sediment nitrogen isotope data illustrates the impact of variable N sources, from fossil fuel combustion, synthetic fertilisers, land use change and wastewater on UK lakes.

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ABSTRACT

Rationale. The global nitrogen cycle has been fundamentally reconfigured by human activity in the last two centuries. This alteration has played out especially in freshwaters, where changes in nitrogen inputs have transformed whole lake and river ecosystems by boosting primary production in nutrient-limited systems and contributing to eutrophication in others. Global alteration of N is manifest in the nitrogen isotopic signature of organic material that moves through freshwater ecosystems. Lake sediments store organic material and can provide a unique historical, stratigraphic record of changes in nitrogen inputs that can be compared with modelled historical changes and contemporary monitoring. **Methodology.** Lake sediment nitrogen isotopes ($\delta^{15}\text{N}$) were measured in archived lake sediment cores ($n = 90$) at the well-resolved time intervals of 1850, 1900 and 1980 and ‘surface’ samples with a median date of 1995, determined by radiometric (^{210}Pb ; ^{137}Cs) dating. Total C, N and $\delta^{13}\text{C}$ were also measured to provide co-variables for assessment. Lake and catchment morphometries and environmental parameters were determined from Open-Access databases. Isotopic changes over time and space were investigated using geospatial and multivariate statistical analyses. **Results.** The difference in $\delta^{15}\text{N}$ values between 1850 and 1980 in upland lakes (mean altitude of catchment (MAC) > 300 m above sea level) are largely negative (median -0.75‰ , s.e. 0.15), whereas lowland lakes with MAC < 300 m show a positive difference (median 0.92‰ , s.e. 0.3). **Discussion.** Regional and local differences in the record of $\delta^{15}\text{N}$ in lake sediments are identified and investigated. Co-measured C, N and $\delta^{13}\text{C}$ and known characteristics of the lakes are used to indicate causes of the recorded differences. Limits to the spatial and temporal sampling approach used are identified and discussed, which include issues with bulk sediment analysis, effects of diagenesis and the operation of multiple, potentially conflicting global and local environmental drivers that become integrated to form lake sediment $\delta^{15}\text{N}$ records.

Keywords: Anthropocene, carbon, $\delta^{13}\text{C}$, $\delta^{15}\text{N}$, isotopes, lakes, nitrogen, ^{210}Pb dating, sediment.

Introduction

Bulk lake sediment $\delta^{15}\text{N}$ values provide an integrated and cumulative response to variations in both catchment, in-lake nitrogen (N) compounds and nutrient conditions (Jones *et al.* 2004). Nitrogen compounds from multiple organic and inorganic sources are incorporated into organic matter and waters in a catchment and then assimilated into littoral, planktonic and benthic ecosystems before burial in lake sediments. The isotopic

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composition of nitrogen compounds are fractionated during transport and in the water column due to multiple biogeochemical processes that affect the ratio of ^{15}N to ^{14}N . For example, when oxygen is limited, microbial denitrification preferentially removes ^{14}N , whereas phytoplankton preferentially uptake ^{14}N over ^{15}N . Both processes lead to the residual nitrate pool becoming enriched in ^{15}N , especially as eutrophication progresses. Measurement of the ratio of N isotopes $^{14}\text{N}/^{15}\text{N}$ ($\delta^{15}\text{N}$) in lake sediments therefore provides complementary data to biological (fossil) and other sedimentary proxies to assess past and current trophic conditions of lakes globally (Herczeg *et al.* 2001; Wolfe *et al.* 2006; Vane *et al.* 2010; Han *et al.* 2023; McCarthy *et al.* 2023). In the absence of long-term monitoring data, sediment $\delta^{15}\text{N}$ profiles provide a valuable spatial and temporal record of past N inputs to lake ecosystems.

Depletion of sediment $\delta^{15}\text{N}$ values from the mid-19th and 20th centuries have been reported for remote lakes in the Northern Hemisphere and ice cores in the Arctic (Holmgren *et al.* 2010; Holtgrieve *et al.* 2011; Heard *et al.* 2014) and signify the unprecedented global alteration of the N cycle due to anthropogenic emissions of reactive nitrogen (Nr) (Galloway and Cowling 2002; Dean *et al.* 2014; Mason *et al.* 2023). Isotopic depletion of Nr in the atmosphere is attributed to several anthropogenic causes: fossil fuel emissions, synthetic fertiliser production (Bateman and Kelly 2007) or an associated shift in partitioning caused by increasing atmospheric acidity (Geng *et al.* 2014). As a result of inter-catchment and inter-lake variation that affects the sequestration, cycling and retention of Nr, and thus fractionation of $^{14}\text{N}/^{15}\text{N}$, the magnitude of the changes in $\delta^{15}\text{N}$ values observed vary. However, the timing of this depletion is largely consistent, having a c. 1850 start and a mid-20th century acceleration, prompting the use of $\delta^{15}\text{N}$ trends as a stratigraphic expression and geological marker of the Anthropocene (Holmgren *et al.* 2010; Wolfe *et al.* 2013; Dean *et al.* 2014; Zalasiewicz *et al.* 2021).

By contrast, positive $\delta^{15}\text{N}$ trends of lake and wetland sediments in lower altitudes and latitudes are generally interpreted in terms of changes from land use, nutrient enrichment and subsequent eutrophication (Hodell and Schelske 1998; Elliott and Brush *et al.* 1999; Lake *et al.* 2001; Vander Zanden *et al.* 2005; Brenner 2006). Previous studies of lake sediment $\delta^{15}\text{N}$ in the British Isles point to an upland–lowland divide; with the technique used to assess the ecological response to Nr deposition in the uplands (Curtis *et al.* 2012) and the effects of productivity changes and wastewater loads in lakes spanning the UK oligotrophic to eutrophic gradient (Rawcliffe *et al.* 2010; Vane *et al.* 2010; Woodward *et al.* 2012). Dissimilarities in sediment $\delta^{15}\text{N}$ values between upland and lowland lakes are not unexpected as there are many reasons to expect differences than just the effects of orographic rainfall and wet deposition (Fowler *et al.* 1988; Dore *et al.* 1992; Metcalfe *et al.* 1999; Curtis *et al.* 2007).

Inconsistent lake sediment $\delta^{15}\text{N}$ trajectories exhibited in cores from lakes at multiple spatial scales highlight the need to understand local and regional trends in sedimentary $\delta^{15}\text{N}$ (McLauchlan *et al.* 2013). This paper explores how the historical and global alteration of the N cycle by human activity is recorded in the N isotope ratio of lake sediments at a national scale, in this case, in ^{210}Pb dated sediment cores from lakes across the geographical and altitudinal range of UK catchments. Systematically measured, national-scale datasets of N isotopic change in lake sediments spanning multiple catchments and waterbody types are rare. In the UK and British Isles, stable N and carbon (C) isotopes have been principally used alongside other high-resolution palaeoecological proxies in dated lake cores, e.g. with pigments and diatoms (McGowan *et al.* 2012; Moorhouse *et al.* 2014) and faecal markers (Vane *et al.* 2010) at a small numbers of sites, or have been measured at a regional scale in surface sediments along with other modern limnological parameters as analogues for past lake conditions (Jones *et al.* 2004; Woodward *et al.* 2012). A further study using high-resolution lake sediment records of $\delta^{15}\text{N}$ from 19 lakes focused solely on remote upland sites in the UK (Curtis *et al.* 2012) using dual C and N isotope analysis. Here, 13 of the 19 mainly oligotrophic lakes demonstrated a timing and widespread alteration of N biogeochemistry comparable to Arctic and alpine lakes elsewhere (Holtgrieve *et al.* 2011; Botrel *et al.* 2014).

Here, we present a preliminary exploration of sediment $\delta^{15}\text{N}$ values from well-resolved time intervals (1850, 1900, 1980 and 1995) from a wider, national-scale selection of sites. These data are a novel compromise combining spatial and temporal differences, with reduced stratigraphic resolution balanced against information from a broader range and number of sites. However, such data remain essential to a better understanding of local and regional historical trends in N budgets and to trace the fate of deposited Nr in terrestrial and aquatic ecosystems (McLauchlan *et al.* 2013; Davies *et al.* 2016; Bell *et al.* 2021).

Methods

Sampling of archived sediments

Lake sediment samples from the University College London (UCL) Environmental Change Research Centre (ECRC) core archive were used. Core selection was an iterative process determined by the existence of a well-resolved ^{210}Pb core chronology and the availability of sufficient dried sediment from the chosen temporally defined intervals (below). The spatial distribution of lakes used in this study reflects the prevalence of lake acidification research conducted by the ECRC on UK upland waters from the 1980s, with continued and subsequent lake research by the group resulting in a broader geographical UK distribution (Fig. 1a) as well as sites with catchments from across a spectrum of sizes and elevations (Fig. 1b, c).

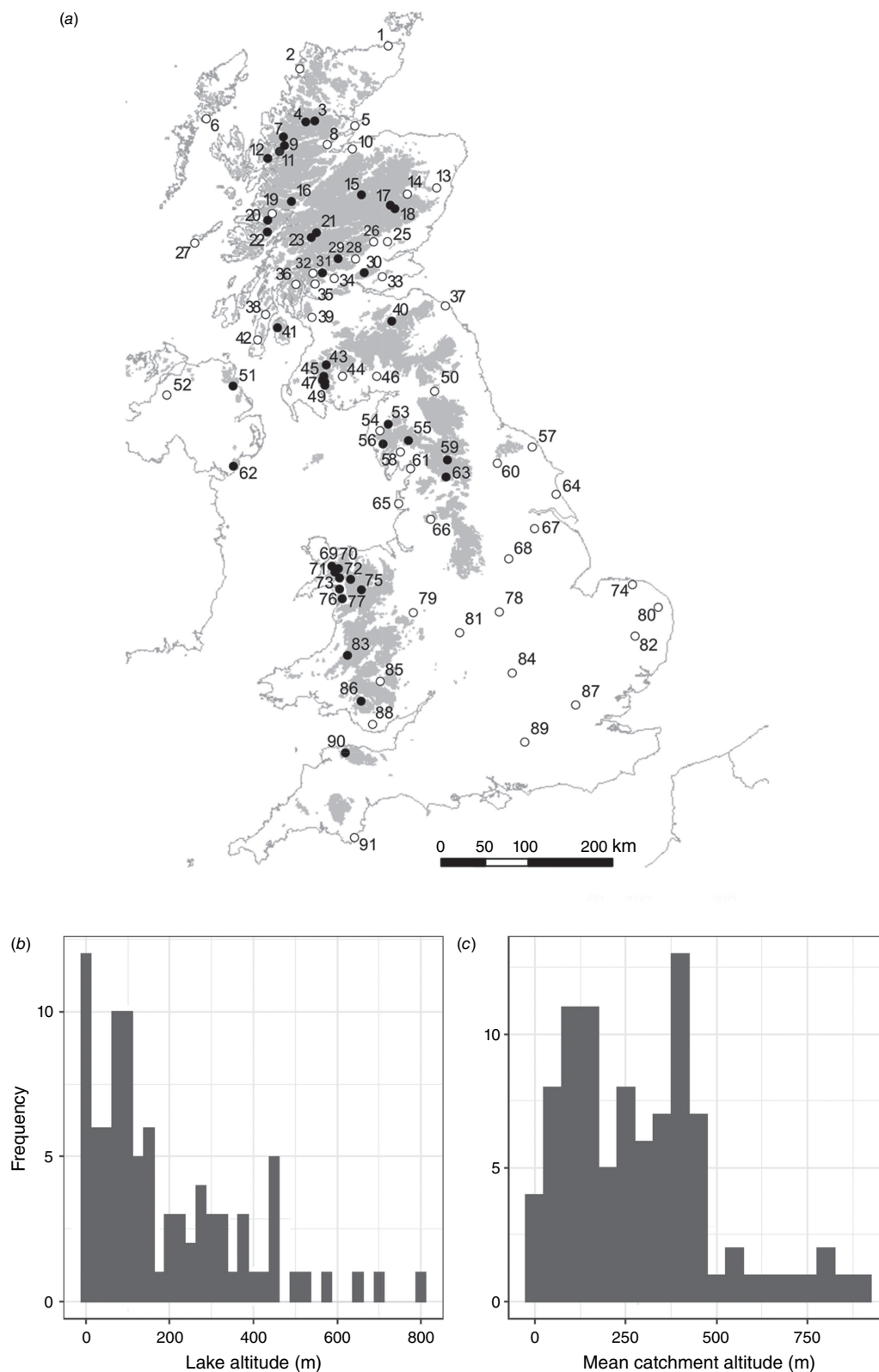


Fig. 1. (a) Locations of lakes and frequency histograms (bins of 50-m altitude) of (b) lake surface altitudes and (c) mean catchment altitudes calculated. Full names of lakes and locations are given in Table S1 of the Supplementary material. Shaded area on map indicates land >300-m altitude. Sites with black circles have mean catchment altitudes (MCA) >300 m.

Sediment samples corresponding to dated intervals of 1850, 1900, and 1980 and 'surface' samples (median date 1995) were selected for stable isotope measurement. A 'top and bottom' approach using limited temporal core data to capture depositional changes in lakes has been used previously and is valid (Bennion *et al.* 2004; Harris *et al.* 2006; Ginn *et al.* 2007), as long as its limits are understood. The top and bottom approach (usually only using two samples) can miss gradual and stepwise changes that have occurred between the past and present and so oversimplifies complex, non-linear processes. Intermediate samples (as in this study) add more detail, but it is a cost v. effort decision between comprehensive analysis of fewer lakes, or as here, using dated sediment intervals from multiple lakes to capture broader landscape-scale changes.

For UK lakes and catchments, and many others globally, the mid-19th century (1850) broadly represents a time prior to or of only limited environmental impacts from fossil fuel emissions, industrialisation and mechanised intensive agriculture (Bell *et al.* 2021). Because of long-term landscape change by human activity, UK lake sediments from 1850 should not be viewed as representing pristine ecological settings (Battarbee *et al.* 2014). However, the choice of 1850 is also pragmatic as for many lake sediment cores analysed since the late 20th century, '1850' often represents the earliest dateable period due to the limits of unsupported ^{210}Pb activity (Appleby 2008).

The year 1900 was selected as the UK population approximately doubled (~20 million to 40 million) between the mid-19th century and this time, concomitant with the growth of large industrial towns and cities and an expanding national rail network fuelled by coal (Church *et al.* 1986; Mosley 2013). From c. 1900, traditional organic soil improvement methods were overtaken by the application of synthesised fertilisers (Brassley 2000). The year 1980 was selected to equate to peak N_r deposition from fossil fuel combustion, which had, by this time, significantly affected UK aquatic systems (Fowler *et al.* 2005) by atmospheric transport and catchment hydrology. Nitrogen fertiliser consumption in the UK also peaked in the 1980s (1987, 1.6×10^6 tonnes) (Department for Environment, Food & Rural Affairs 2022). The age of recent surface sediments in the database range correspondingly to cores collected between 1981 and 2010 (median 1995). Sediments representing the 'surface' were consistently taken from the adjacent depth interval below the surface for analysis to lessen the influence of fresh organic matter and loss of N due to rapid remineralisation with burial (Galman *et al.* 2008, 2009).

^{210}Pb dating

All archived cores were ^{210}Pb dated by direct gamma assay at the Environmental Radiometric Facility at UCL or Liverpool University Environmental Radioactivity Research Centre, using well-type coaxial low-background intrinsic germanium detectors. ^{210}Pb was determined from its gamma emissions at 46.5 keV and ^{226}Ra by the 295 and 352 keV gamma rays

emitted by its daughter isotope ^{214}Pb following 3-weeks storage in sealed containers to allow radioactive equilibration (Appleby and Oldfield 1978). Corrections were made for the effect of self-absorption of low-energy gamma rays within the sample. Errors of ^{210}Pb dates increase with age due to the half-life of unsupported ^{210}Pb (22.3 years); typical errors for 1850–1900 are in the order of ± 15 –25 years and 1980–present ± 2 years. Further details on the core chronologies are provided in Supplementary Table S1.

Nitrogen and carbon isotope measurements

Sediments were analysed simultaneously for $^{15}\text{N}/^{14}\text{N}$ and $^{13}\text{C}/^{12}\text{C}$, as well as total N and C, at the UC Davis Stable Isotope Facility (Davis, CA, USA) using isotope ratio mass spectrometry on Hydra 20–20 or Anca-GSL isotope ratio mass spectrometers.

Bulk sediment was homogenised by a stainless-steel ball mill (Retsch MM200) from which ~10–12 mg of sediment was encapsulated in 9- × 5-mm tin capsules (Elemental Microanalysis) prior to combustion. Samples were interspersed during analysis with several replicates of laboratory standards. Elemental concentrations and stable isotope compositions were co-measured with nylon, bovine liver (SRM1577a), peach leaves (SRM1547) and glutamic acid (USGS-41) reference materials. Sample values were finalised by correcting the values for the entire batch of samples based on the known values of the included laboratory reference material (see <https://stableisotopefacility.ucdavis.edu/carbon-and-nitrogen-solids>).

The isotopic ratios of $^{15}\text{N}/^{14}\text{N}$ and $^{13}\text{C}/^{12}\text{C}$ are expressed using the delta (δ) notation, using the following equation:

$$[(R_{\text{sample}} \div R_{\text{standard}}) - 1] \times 1000$$

where R is the $^{15}\text{N}/^{14}\text{N}$ or $^{13}\text{C}/^{12}\text{C}$ ratio in the measured sample or standard. The standard for N isotopes is the $\delta^{15}\text{N}$ of atmospheric N (commonly referred to as AIR), and for $\delta^{13}\text{C}$ the standard is Vienna Pee Dee Belemnite (VPDB).

As the aim of the study was to compare N isotopic changes, further bias was not introduced with the acidification of samples to remove inorganic C (Harris *et al.* 2001; Meyers and Teranes 2001). Inorganic C in lake sediments comes from water chemistry (dissolved inorganic carbon) and geological-scale carbon fractionation (e.g. carbonate minerals). Inorganic carbon typically has higher $\delta^{13}\text{C}$ values due to equilibrium fractionation during carbonate precipitation dissolution. Organic carbon has lower $\delta^{13}\text{C}$ values due to biological discrimination against ^{13}C . The presence of inorganic C in lake sediments therefore limits the use of $\delta^{13}\text{C}$ as a proxy for changes in organic activity in the lake and catchment. This limitation is recognised in this paper where $\delta^{13}\text{C}$ values are used to interpret $\delta^{15}\text{N}$ changes. $\delta^{15}\text{N}$ is often measured on untreated lake sediments, for example, in upland oligotrophic lake systems where inorganic C and N

sources are considered negligible (Jones *et al.* 2004; Battarbee *et al.* 2014; Moorhouse *et al.* 2014). Acidification treatments can also show significant non-linear patterns on measured $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values, including no difference between different methods (Brodie *et al.* 2011).

GIS and other data sources

Lake and catchment morphometries were determined from Ordnance Survey Open Data Land-Form PANORAMA raster data (50- × 50-m cell) using ArcGIS (ver. 3.0, see <https://www.arcgis.com/>; Hughes *et al.* 2004). Lake codes used in this paper correspond to ECRC sediment core codes assigned at the time of collection from the lakes. Lake names and associated codes are shown in Table S1.

Modern climate parameters (mean annual temperature and precipitation) were calculated in ArcGIS for each catchment polygon based on 5-km gridded data from UKCP09 (UK Met Office) data. Population density values for the lake catchment polygons were calculated for England and Wales (Office National Statistics 2011). Population data for Scotland came from the 2001 Census Output Area data set (National Records of Scotland 2001). Modern climate and population data are used here to compare with sediment $\delta^{15}\text{N}$ values. Human Footprint map data (1993 values) were used to compare UK lake sediment $\delta^{15}\text{N}$ with cumulative indicators of human impact (McLauchlan *et al.* 2013; Venter *et al.* 2016). ArcGIS (Spatial Statistics Toolbox) was used to calculate spatial autocorrelation (Global Moran's I) of $\delta^{15}\text{N}$ values. Z-score (standard deviation) and P -value (probability) from the autocorrelation calculation provide a numerical assessment of significant clustering or dispersion and distance from a random process. Generalised additive (mixed) models (GAMs) and generalised additive models for location, scale and shape (GAMLSS) were estimated using the R (ver. 3.4.4, R Foundation for Statistical Computing, Vienna, Austria, see <https://www.r-project.org/>) package *mgcv* (ver. 1.8-25, see <https://cran.r-project.org/package=mgcv/>; Wood *et al.* 2016; Wood 2017). Isotope data are available Open Access from the NERC Environmental Information Data Centre (EIDC) (Turner and Rose 2015).

Results

Catchment–lake physiography

Lakes with a surface area greater than 1 ha are found from sea level to 1215 m above sea level in the UK (UK Lakes Portal, see <https://eip.ceh.ac.uk/apps/lakes/>) but lakes with water surface altitudes lower than 300 m dominate (70%) our dataset ($N = 90$; Fig. 1b). Mean catchment altitude (MCA) was considered to provide a more inclusive assessment of potential changing N sources to lakes in the UK, i.e. water bodies at the base of steeply glaciated catchments in Scotland and Wales. The dataset is largely split between lakes with an MCA less or greater than 300 m ($n = 48, 42$ respectively; Fig. 1c). In the UK, land above 300 m (~1000 ft) altitude is broadly considered 'upland' with a cooler and wetter climate found in the north and west, compared to the warmer, drier 'lowland' in the south and east (Averis 2004). This loose upland–lowland altitudinal divide in the UK is also marked by notable differences in vegetation, geology and historical land-use. Upland lakes in the UK are usually more remote, low-nutrient systems, and sensitive to acid deposition due to thin soils and underlying geology with little buffering capacity. Lowland aquatic systems in the UK have a long history of being affected by far more intense land-use and subsequent nutrient enrichment and eutrophication, as a consequence of being downstream and proximal to large populations. This lowland–upland division, therefore, provides a useful framing to observe how contrasting lake types have responded to human activity and global and national changes in historical atmospheric and nutrient N inputs.

Spatial and temporal changes of $\delta^{15}\text{N}$

Sediment $\delta^{15}\text{N}$ values from all locations, ages and depths vary between -1.6 and 13.3 , although lakes with MCA > 300 m do not exceed 5.6 . Lowland lake sediments deposited in 1850 exhibit $\delta^{15}\text{N}$ values comparable to upland sites. Spatial autocorrelation values (Table 1) show that the 1850 $\delta^{15}\text{N}$ pattern is less spatially structured, representing pre-industrial, geographically dispersed and variable N inputs into lakes. The 1900 z -score ($z = 2.18, P = 0.029$) indicates

Table 1. Global Moran's I spatial autocorrelation of $\delta^{15}\text{N}$ values.

	1850 $\delta^{15}\text{N}$	1900 $\delta^{15}\text{N}$	1980 $\delta^{15}\text{N}$	1995 $\delta^{15}\text{N}$
Moran's Index	0.024	0.126	0.180	0.147
Variance	0.004	0.004	0.004	0.004
z -score	0.556	2.181	2.919	2.384
P -value	0.578	0.029	0.004	0.017
Comments	Not significantly different than random	<5% likelihood clustered pattern is result of random chance	<1% likelihood clustered pattern is result of random chance	<5% likelihood clustered pattern is result of random chance

Conceptualisation: inverse-distance. Distance method: Euclidean. Row standardisation: true

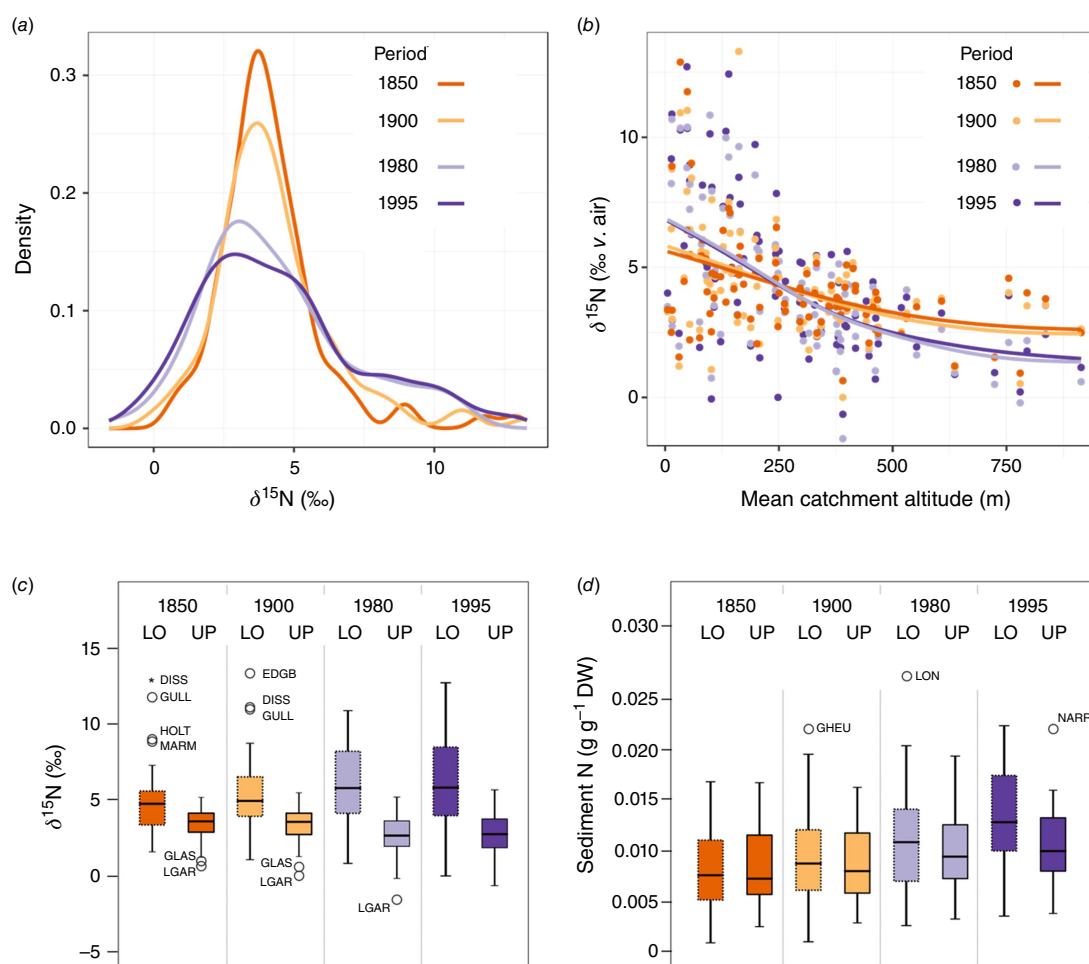


Fig. 2. Lake sediment $\delta^{15}\text{N}$ values at 1850, 1900, 1980 and present (median 1995). (a) Frequency distribution. (b) GAMLSS curved fits between sediment $\delta^{15}\text{N}$ values and mean catchment altitude. (c) Box and whisker plot comparing lowland (LO) and upland (UP) sediment $\delta^{15}\text{N}$ values and (d) comparing sediment N. See Fig. 1 for locations and names of outliers.

a more spatially clustered pattern than the 1980 value ($z = 2.91$, $P = 0.035$), with a $<1\%$ likelihood that $\delta^{15}\text{N}$ clustering is random, indicating more spatially structured N drivers.

The frequency distribution of sediment $\delta^{15}\text{N}$ values in UK lake sediments becomes more heavily tailed (Fig. 2a) between 1850 and 1995, with the greatest difference between 1900 and 1980. Highly significant GAMLSS fits, specifically in relation to altitude (Fig. 2b), also show little difference between 1850–1900 and 1980–1995. This switch illustrates the known 20th Century disruption to the global N-cycle, resulting in higher $\delta^{15}\text{N}$ values in lowland systems and lower $\delta^{15}\text{N}$ values in upland lakes.

Extreme low $\delta^{15}\text{N}$ values in the upland group from 1850 include Loughgarve (LGAR) in Northern Ireland and Llyn Glas (GLAS) in Yr Wyddfa, Wales. LGAR is a shallow (maximum depth 0.9 m) moorland lake with an upland bog catchment and is, or likely was, N-limited (Gibson *et al.* 1995), whereas at GLAS, low sediment $\delta^{15}\text{N}$ values also suggest

nutrient limitation due to a high flushing rate, lack of sewage and fertiliser and organic matter dominantly from lacustrine sources and low $\delta^{15}\text{N}$ catchment vegetation (Jones *et al.* 2004). These lakes are observed as having outlier N isotope values in both upland and lowland groups up to the present. The relationship between bulk sediment total N and $\delta^{15}\text{N}$ for all time interval samples from all 90 UK lake cores is poor ($r^2 = 0.007$, $P < 0.1$), although an increase since 1850 in total N in lowland lakes is apparent (Fig. 2d).

Comparison of past and current sediment $\delta^{15}\text{N}$ changes with co-measured C isotopes ($\delta^{13}\text{C}$) and the ratio of C:N provide more detail on temporal changes. Both clustering and dispersal of values over time are observed in plots of dual C:N isotope values (Fig. 3a); a negative $\delta^{15}\text{N}$ shift of upland sites is apparent.

High $\delta^{13}\text{C}$ values are evident in more alkaline lakes, i.e. Hornsea Mere (HORN), Loch a Phuill (PHUI), Hawes Water (HAWE), Malham Tarn (MALH) and Upton Broad (UPTON) with significant inorganic C. These lakes have been

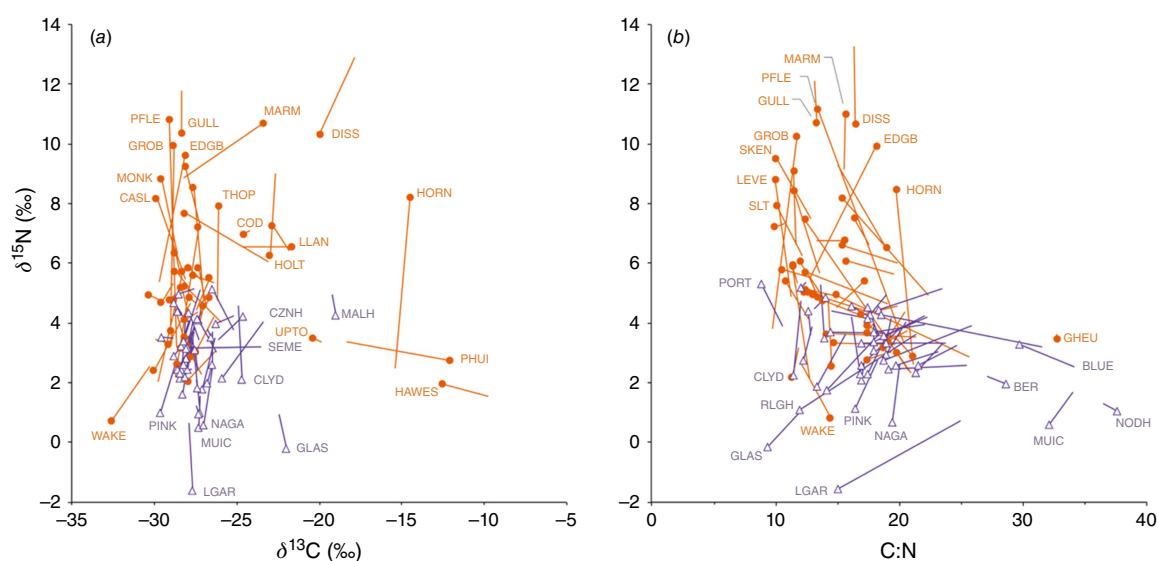


Fig. 3. Dot and tail trajectory plots of (a) sediment $\delta^{15}\text{N}$ v. $\delta^{13}\text{C}$ and (b) sediment $\delta^{15}\text{N}$ v. C:N. Dots indicate 1980, the end of tails indicate 1850, circles indicate MCA < 300 m, and triangles indicate MCA > 300 m. Codes for lakes are in Table S1.

historically affected by eutrophication (Bennion *et al.* 2004; Ayres *et al.* 2007; Rawcliffe *et al.* 2010; May *et al.* 2011) and so phytoplankton photosynthetic fractionation may have mediated sediment $\delta^{13}\text{C}$ values.

Sediments with lower C:N values in 1980 have a greater range of $\delta^{15}\text{N}$ values than lake sediments with a higher contribution from allocthonous (terrestrial) sources indicated by high C:N values (Fig. 3b). Terrestrial organic C inputs to lakes with low and negative trending sediment $\delta^{15}\text{N}$ are seen in upland, peat-abundant catchments, e.g. Llyn Berwyn (BER), Loch Muick (MUIC), Loch na Gabhalach Nodha (NODH) and Grey Heugh Slack (GHEU). A negative shift in both sediment $\delta^{15}\text{N}$ and C:N has occurred in upland sites of GLAS, LGAR, MUIC and Llyn Clyd (CLYD).

Because of the age range of the surface samples (range 1981–2012, median 1995) and potential effects of sediment–water interface diagenesis, the difference between 1850 and 1980 values is here used to show the UK-scale change in lake sediment $\delta^{15}\text{N}$ trajectories during a period of unprecedented industrial and land-use change (Fig. 4). The spatial distribution of changes shows both negative and positive historical trends. Negative trends are concentrated in upland areas of Scotland, Wales, Northern Ireland, the Lake District and northern England. Negligible to more positive differences occur scattered across the UK but noticeably in more lowland areas, which are the inverse of upland areas being warmer and with a higher population density (Fig. 4 and 5). The greatest positive differences (Table 2) in sediment $\delta^{15}\text{N}$ are observed in shallow, eutrophic lakes with historical nutrient inputs from urban and agricultural sources (Sayer *et al.* 1999; Bennion *et al.* 2004, 2009; May *et al.* 2011; Gunn *et al.* 2013).

Comparing $\delta^{15}\text{N}$ values with altitude indicates an effect of near sea-level lakes in high glaciated catchments. Positive value changes to the present are seen in the higher mean catchment altitudes of Loch Lomond (LOMO), Loch Maree (MARE) and Loch Shiel (SHIE) in Scotland. Nutrient enrichment is suggested to have caused positive sediment $\delta^{15}\text{N}$ shifts in the uplands (e.g. in MALH and Llyn Fach, FACH), but above 600 m all sites show a negative change (Fig. 5b).

Magnitude and trajectory of change can also be determined by a best-fit regression between time intervals and $\delta^{15}\text{N}$ values for each of the lake cores; the slope values representing rate of change per year (year^{-1}) and direction ($\pm$) of $\delta^{15}\text{N}$ change (Fig. S1 of the Supplementary material). This linear representation of change is only a guide. Mapped trajectories from four-point regressions (Fig. S1), show a similar spatial pattern as the difference in $\delta^{15}\text{N}$ values, but here also provide rate and direction of change information.

Discussion

Changes in sediment $\delta^{15}\text{N}$ recorded in UK lakes are clearly a complex function of catchment and lake integration (Curtis *et al.* 2007). The spatial and temporal dataset of sediment $\delta^{15}\text{N}$ values presented here shows that global-scale historical N changes can be recognised, but considerable differences occur due to physiographic and anthropogenic factors and the biogeochemical complexity of soils, organic matter, in-lake and post-burial processes (Meyers and Teranes 2001; Brahney *et al.* 2014). However, while recognising this complexity, it is also clear that despite using a low temporal resolution, opportunistic and iterative sampling process of

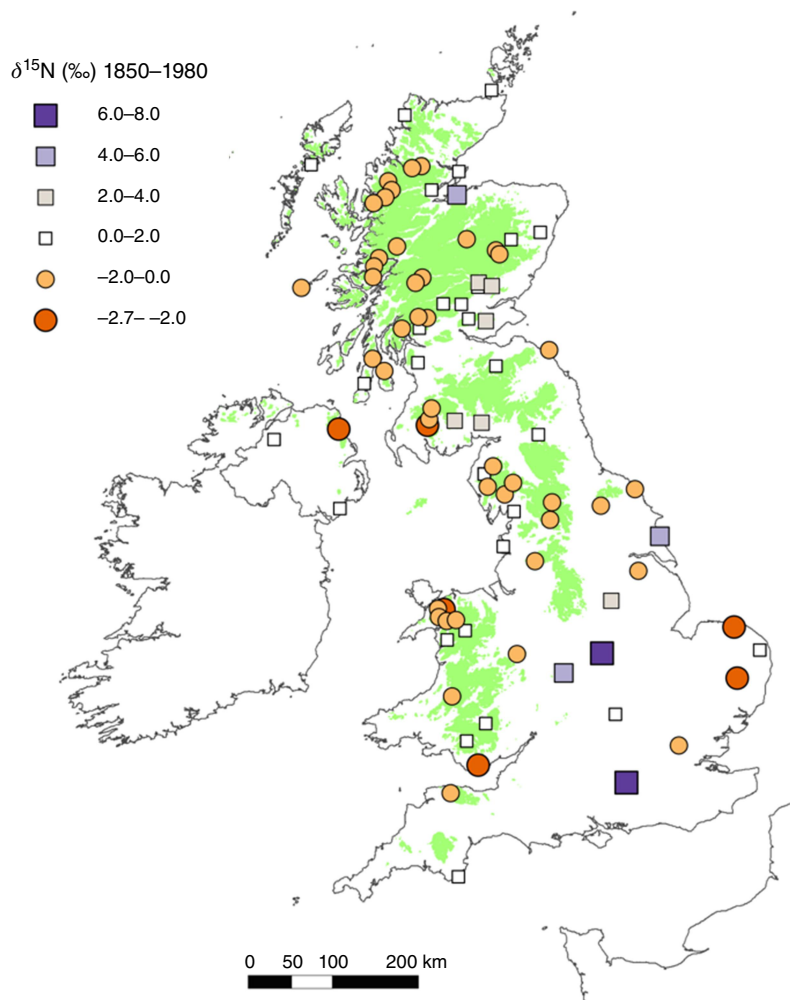


Fig. 4. Difference in lake sediment $\delta^{15}\text{N}$ between 1850 and 1980. Positive values indicate a positive change in $\delta^{15}\text{N}$ from 1850 to 1980. Light green shading is UK land >300 m above sea level.

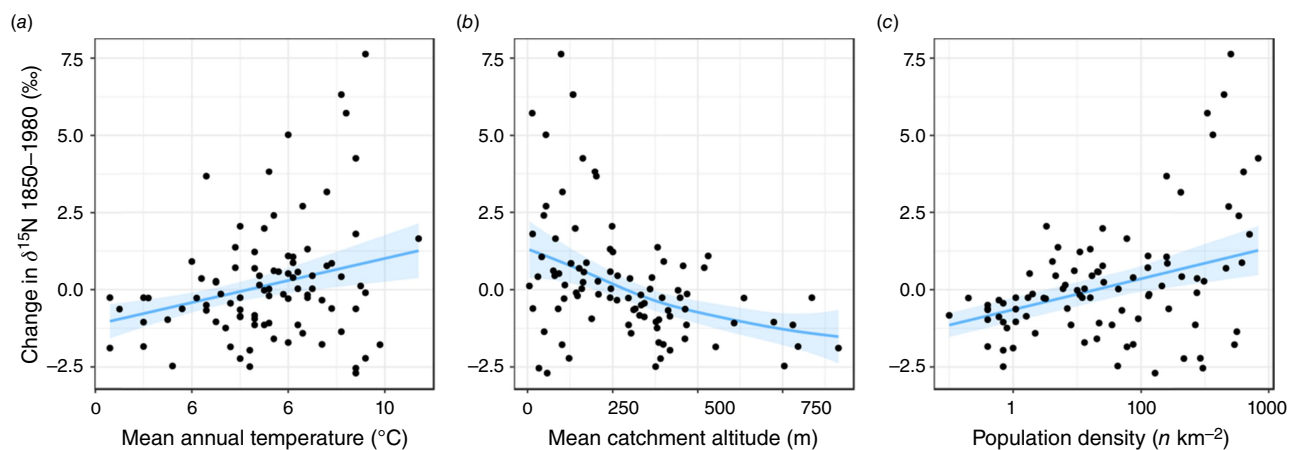


Fig. 5. Change in UK lake sediment $\delta^{15}\text{N}$ from 1850 to 1980 in relation to environmental parameters: (a) mean annual temperature, (b) mean catchment altitude and (c) population density.

available archived lake sediments, the dataset validates and communicates very well the distinct ecosystem change in water bodies across upland and lowland Britain that has occurred due to unprecedented human activity.

Our data here show that sedimentary $\delta^{15}\text{N}$ values should always be first considered on an individual core basis, within a local and regional context, as knowledge of the unique isotopic signature of the N inputs into lakes is critical

Table 2. UK water bodies measured with the greatest positive and negative difference in sediment $\delta^{15}\text{N}$ between 1850 and 1980.

$\delta^{15}\text{N}$ Change 1850 to 1980	Code	Water body name and UK location	MCA (m above sea level)	Population ($n \text{ km}^{-2}$)
Positive shift (history of eutrophication)				
+7.6	PFLE	Fleet Pond, Hampshire	99	2250
+6.3	GROB	Grobby Pool, Leicestershire	133	1998
+5.7	HORN	Hornsea Mere, Yorkshire	14	1094
+5.0	FLEM	Loch Flemington, Inverness	54	1336
+4.2	EDGB	Edgbaston Pool, Birmingham	162	6840
+3.8	LEVE	Loch Leven, Perth & Kinross	197	4036
Negative shift (atmospheric deposition)				
-1.85	LLAG	Llyn Llagi, Snowdonia	552	60
-1.9	NARR	Loch Narroch, Galloway	418	0.7
-1.9	NAG	Lochnagar, Cairngorms	914	1
-2.2	LGAR	Loughgarve, Northern Ireland	395	470
-2.5	RLGH	Round Loch of Glenhead, Galloway	376	0.7
-2.5	CLYD	Llyn Clyd, Snowdonia	755	43
Negative (recovery from historical eutrophication – wastewater)				
-2.5	DISS	Diss Mere, Suffolk	33.4	930
-2.7	HOLT	Holt Hall Lake, Norfolk	57.8	165

Grouped by known lake histories.

(Botrel *et al.* 2014). Physiographic and anthropogenic influences have clearly influenced sediment $\delta^{15}\text{N}$ trends (Fig. 5, 6); with cool, high-elevation (MCA > 300 m), oligotrophic lakes showing negative $\delta^{15}\text{N}$ shifts (up to -2.5) (Table 2), and warmer, low-elevation lakes showing far greater increases in sedimentary $\delta^{15}\text{N}$ values (maximum 7.6). Low values of both $\delta^{15}\text{N}$ and C:N point to sediments in upland lakes (e.g. GLAS, LGAR, Round Loch of Glenhead (RLGH) and CLYD) having accumulated atmospheric-N directly by in-lake stimulation of autochthonous organic matter (lacustrine algae and plankton) (Jones *et al.* 2004).

In lower altitude lakes, the $\delta^{15}\text{N}$ history of eutrophication is not so uni-directional due to the variety and lesser or greater amount of N inputs in the recent historical past. For example, DISS, Marton Mere (MARM) and HOLTU received sewerage and wastewater inputs before 1850 (Peglar 1993; Yang 2010; Turner *et al.* 2013). Marine-derived N is proposed for the observed pattern seen at Gull Ponds (GULL) that was known for its large pre-World War 2 seagull colony (Hollom 1940) as well as the near-coast MARM, which is likely to have had similar guano inputs to other UK lakes with gull colonies, such as Hickling Broad (Irvine *et al.* 1993). A marine-guano $\delta^{15}\text{N}$ sediment signal is also suggested to occur in near-coast,

wildfowl-rich lakes Monk Myre (MONK), Coldingham Loch (COD) and Loch Skene (SKEN) prior to fertiliser-induced eutrophication. Increased $\delta^{15}\text{N}$ values certainly correspond with early and increasing 20th Century eutrophication and hyper-eutrophication due to wastewater inputs at EDGB, Thoresby Lake (THOP), Fleet Pond (PFLE) and Hornsea Mere (HORN) (Turner *et al.* 2013; Bennion *et al.* 2018).

Due to their geographical location and relative isolation, the negative shifts seen in small, woodland lakes Psygodlyn Mawr (PYSG) in South Wales and WAKE (Epping Forest, London) suggest the influence of atmospheric deposition of fossil fuel N from industrial and urban areas (Fowler *et al.* 2005). Supporting evidence from other indicators of fossil fuel atmospheric deposition (in WAKE, spheroidal carbonaceous particles; Turner *et al.* 2013) needs, however, to be considered against the inputs from the range of $\delta^{15}\text{N}$ values found in woodland soils, trees and leaf litter (Falxa-Raymond *et al.* 2014) as well as sedimentation in anoxic, tree-sheltered waterbodies. Low values of both $\delta^{15}\text{N}$ and $\delta^{13}\text{C}$ in WAKE suggest the selective preservation of ^{13}C -depleted compounds and bacterial growth adding ^{15}N -depleted biomass to the bulk sediment (Lehmann *et al.* 2002).

Although a lack of a globally consistent trend in sedimentary $\delta^{15}\text{N}$ is reported (McLauchlan *et al.* 2013) due to local

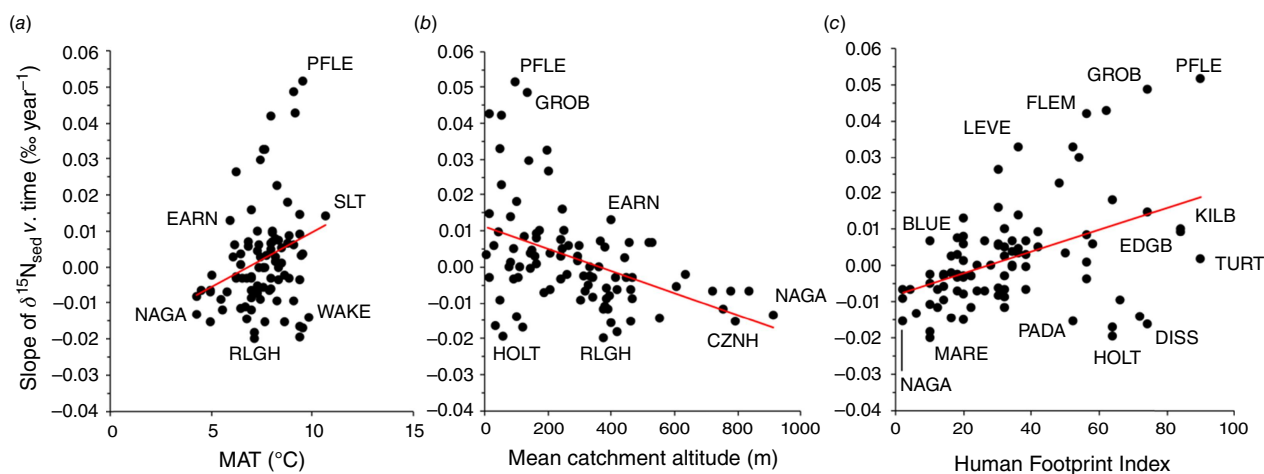


Fig. 6. UK Lake sediment $\delta^{15}\text{N}$ trajectories from 1850 to 1995. The y-axis is the slope of $\delta^{15}\text{N}$ v. time (‰ year^{-1}) from four-point linear regressions at individual sites (after [McLauchlan et al. 2013](#)). Lake trajectories in relation to (a) mean annual temperature (MAT) ($r^2 = 0.0775$, $P < 0.01$), (b) mean catchment altitude (MCA) ($r^2 = 0.188$, $P < 0.001$) and (c) Human Footprint Index (1993 values) from [Venter et al. \(2016\)](#) ($r^2 = 0.190$, $P < 0.001$). For site codes see Table S1.

conditions, a national-scale trend in the UK since 1850 identified in this study indicates an increase in sedimentary $\delta^{15}\text{N}$ values due to population density (Fig. 5c) and ‘Human Footprint’ ([Venter et al. 2016](#)) (Fig. 6c).

Other considerations

Consistency of the $\delta^{15}\text{N}$ sediment record can be investigated by measurement of multiple dated cores in the same lake ([Schelske and Hodell 1995](#); [Herczeg et al. 2001](#)) and uncertainty reduced by measurement of trap sediments, seston and within-lake ecological compartments ([Owen et al. 1999](#); [Jones et al. 2004](#)). Similarly, increasing the number of replicates representing a particular time interval, or indeed increasing the stratigraphic resolution, would improve confidence, but adds a significant workload to stratigraphic studies featuring large numbers of sites. This study was designed to contribute to the Long Term Large Scale (LTLS) Macronutrient Model for the NERC Macronutrient Cycles programme (2014–2017) that sets out to simulate terrestrial and freshwater fluxes of nutrients at a broad scale (multiple catchments) over decades and centuries, rather than lake-specific changes. This compromise will be a continued feature of large-scale research always requiring more detail spatially and temporally.

Furthermore, the bulk measurement of lake sediments, by their integrative nature, homogenise C and N from a broad range of organic sources, which is a significant source of uncertainty in $\delta^{15}\text{N}$ measurements. Small changes in organic composition must be considered with the mass spectrometry method, for example, the effect of microscopic tree leaf particles in an algal-dominated sediment sample. $\delta^{15}\text{N}$ measurements on cladocera remains ([Perga 2010, 2011](#)), fish scales ([Ventura and Jeppesen 2010](#)) and specific compounds

in lake sediments ([Enders et al. 2008](#); [Tyler et al. 2010](#)) clearly provide more certainty than bulk sediment but add an extra level of sampling effort at the least. Because of the link between C and N in lake sediments, it is clearly advisable to increase analytical effort and replicate dual C:N isotope measurements on bulk samples that have been acidified, to reduce some of the uncertainty from inorganic carbon (especially when interpreting measurements across diverse geological landscapes).

Improved understanding, or at least greater recognition, of the effect of early diagenesis on core profiles should reduce the ambiguity of what stable isotope values represent in bulk lake sediment samples. Post-depositional processes change N and C compounds, and sediment records of their isotopes also reflect this ([Lehmann et al. 2002](#); [Brahney et al. 2014](#)). Understanding how important diagenetic processes are is problematic as local redox conditions and organic matter availability, such as the spatial variability of pre-burial controls, will influence the extent of change in $\delta^{15}\text{N}$ values. With such potential variability, the consistent decline in $\delta^{15}\text{N}$ during the last 150 years measured in UK high and remote lakes nonetheless supports causality due to changes in atmospheric N inputs rather than diagenesis. In conjunction, this atmospheric process (depletion of $\delta^{15}\text{N}$ values) must have also reduced the more positive $\delta^{15}\text{N}$ values in lowland lakes, when in receipt of comparable atmospheric N inputs.

This exploratory low temporal resolution analysis clearly shows a demand for more specific and detailed analytical and modelling techniques to disentangle deposition and post-depositional effects on N isotope patterns ([Galman et al. 2008, 2009](#)), which needs to be resolved across a broader range of lake types and timescales. One clear aspect to resolve is the effect that diagenesis has on the $\delta^{15}\text{N}$ signal over decadal and century scales as organic matter is buried

and preserved in different lakes. A result of how long the ECRC has been working on UK lakes indicates one area for future study, i.e. the repeat measurement of dated sediments collected at different times from the same lake. Cores from Loch Shiel (SHIE2 and SHIE5) were collected in 1995 and 2006 and analysed as part of this research. Applying a null modelling approach (Brahney *et al.* 2014), the effect of diagenesis on $\delta^{15}\text{N}$ values is inconclusive, with a divergence of values occurring in sediments at present and also in 1850 (Fig. S2 of the Supplementary material). Little difference in $\delta^{15}\text{N}$ values from 1900 was observed, and from the 1980 layer that had an extra 11 years in the lake, the $\delta^{15}\text{N}$ difference was only a value of 0.2.

Although the range of environmental processes and variables that determine bulk lake sediment $\delta^{15}\text{N}$ values are broad and in need of further scrutiny, $\delta^{15}\text{N}$ values reported in this study nonetheless show a transformation of UK lakes by human activity in the last 150 years. An interesting consideration for future study would be to analyse these lakes again now and in 2050 to capture a full two centuries of impact and potential recovery, as well as potential issues of the preservation of the $\delta^{15}\text{N}$ record in lake and archived materials. Significant efforts have been made since the late 20th Century to reduce N pollution; most directly through fossil fuel emission reductions and improved wastewater treatment, however inputs into aquatic systems due to residual nitrogen stored in soils and groundwater will continue to influence ecosystems for decades into the future.

Conclusion

Both global and local patterns and processes have determined the trajectory of changing sediment $\delta^{15}\text{N}$ trends in UK lakes since 1850. Sedimentary $\delta^{15}\text{N}$ trends show that the technique is sensitive to measure regional physiographic and anthropogenic influences, not just global controls of environmental parameters (McLauchlan *et al.* 2013).

The unprecedented transformation of the global N cycle due to fossil fuel combustion and release of artificial fertilisers and wastewater into freshwaters is recognised in UK lake sediment records of $\delta^{15}\text{N}$. Because of the historical integration of N sources, a sediment $\delta^{15}\text{N}$ decrease, attributed to fossil fuel combustion and atmospheric deposition of Nr is most apparent in upland lakes with mean catchment altitudes above 300 m. Sediment $\delta^{15}\text{N}$ values in lowland UK lakes show that anthropogenic changes to N cycling also occurred simultaneously due to atmospheric and terrestrial biogeochemical changes and wastewater inputs; with modern sediment $\delta^{15}\text{N}$ values in some lakes showing amelioration or recovery from past nutrient loads. This integration of $\delta^{15}\text{N}$ signals is a key reason why other palaeoecological techniques are essential to understand sources and drivers.

Determining the precise environmental drivers and timing of sediment $\delta^{15}\text{N}$ values requires lake by lake and

individual core analysis, but when viewed at a UK scale, the variability and trajectory of sediment $\delta^{15}\text{N}$ changes are evidently associated with elevation (250–300 m) above sea level, which in the UK is synonymous with a distinct biogeochemistry, population density, climate and nature and intensity of catchment land-use.

$\delta^{15}\text{N}$ core profiles reflect these environmental gradients, but because of the integrating nature of lake sediment and the bulk nature of sampling, disentangling cause from effect on down-core changes is not straightforward, and evidence from sediment $\delta^{15}\text{N}$ requires support from site specific historical information (for example documented ecological, physical changes) and from other palaeoenvironmental proxies (e.g. palaeoecology, sediment environmental DNA).

Adherence to and the reporting of methodological procedures used is clearly important for continued use of the $\delta^{15}\text{N}$ sediment technique, as well as access to data from previous studies. Considering the effects of using bulk sediment, different methods of removal of inorganic and organic components, sample sizes and mass spectrometers, there is a continued need for standardising sample preparation, use of standards, analysis with other palaeoecological techniques and ring-testing of values generated between laboratories to reduce uncertainty and improve pan-regional $\delta^{15}\text{N}$ sediment research.

Supplementary material

Supplementary material is available [online](#).

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Data availability. The sediment, isotope and location data used in this paper are available as Open Access from the NERC Environmental Information Data Centre (EIDC) archive (Turner and Rose 2015).

Conflicts of interest. The authors declare that they have no conflicts of interest.

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